Project Title
Résumé–Job Matching and Talent-Marketplace Engine

Project Report submitted for the partial fulfilment of grade for the course

Data Structures and Algorithms -3 (25CS2103E)

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ABSTRACT

The Résumé–Job Matching and Talent-Marketplace Engine is designed to improve the recruitment process by automatically matching candidates with suitable job opportunities. Traditional hiring methods require recruiters to manually review large numbers of résumés, which can be time-consuming, inconsistent and inefficient. The proposed system indexes résumés and job descriptions, extracts important information such as skills, qualifications, experience and job requirements, and calculates a compatibility score between each candidate and available role.

The system normalises different terms used to represent similar skills, allowing expressions such as "machine learning," "ML" and "predictive modelling" to be recognised as related competencies. Candidates are then ranked according to their relevance to a particular job. In addition to individual matching, the engine uses an optimal-assignment approach to assign candidates to multiple roles while avoiding duplicate or inefficient allocations. It also identifies the minimum set of skills required to form a team for a given project or organisational requirement.

The project applies data structures and algorithms such as hash maps, inverted indexes, string-matching techniques, priority queues, graphs, matching algorithms and set-cover approaches. The final system provides recruiters with ranked candidate recommendations, skill-gap information, suitable job suggestions and transparent explanations for every match. This reduces recruitment effort, improves consistency and supports faster and more informed hiring decisions.

Signature of the Guide